

Vaccination Cohort & Dose Scheduling System

Requirements Table (SRS Format)

Problem Statement #17 | Healthcare & Telemedicine: Vaccination Cohort & Dose Scheduling System. Five Functional Requirements (FR-001–FR-005) and two Non-Functional Requirements (NFR-001–NFR-002), each with Req ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enforce dose interval rules (minimum 28 days after Dose 1) before unlocking Dose 2 booking for a citizen.	High	Pass: Dose 2 booking is blocked if interval < 28 days. Fail: an early vaccination slot is confirmed.	Prevents medically unsafe early re-dosing (given requirement).
FR-002	Functional	The system shall allow a citizen to register with demographic details and be automatically assigned to an eligible vaccination cohort.	High	When a citizen submits age and category details, the system assigns the correct cohort and displays the eligibility date.	Ensures rollout is prioritized correctly by demographic group.
FR-003	Functional	The system shall allow a citizen to book an available slot at a vaccination center, decrementing that center's capacity in real time.	High	When a slot is booked, the center's remaining capacity decreases by exactly 1 and the slot is removed from availability once capacity reaches 0.	Prevents overbooking of physical vaccination centers.
FR-004	Functional	The system shall allow a Vaccination Officer to mark a dose as administered against a citizen's booking record.	High	When the officer confirms administration, the booking status changes to "Completed" with a timestamp within 2 seconds.	Creates the authoritative record that certificate generation depends on.
FR-005	Functional	The system shall automatically generate a digitally signed QR vaccination certificate once a dose is marked as administered.	High	The QR certificate is available for citizen download within 5 seconds of dose confirmation.	Enables immediate, verifiable proof of vaccination for the citizen.
NFR-001	Nonfunctional – Performance & Security	The vaccination certificate verification endpoint shall authenticate digitally signed QR codes offline/online in under 150 ms.	High	Pass: benchmarking tests confirm target latency and security standards under simulated peak load.	Fast, secure verification is required at high-traffic checkpoints (given requirement).
NFR-002	Nonfunctional – Reliability & Scalability	The system shall prevent double-booking of the same vaccination slot under concurrent booking requests.	High	A load test with 500 simultaneous requests for a single slot results in exactly 1 confirmed booking and 499 rejections.	Data integrity is critical for a public health scheduling system.